



Business Case for The Qubit

by
KM5A

Jiali Gu
Ze Kai Chong
Hermanto
Tiantian Wang
Mohon Dash Tanu
Haoyu Zhang

Project Description

The Qubit is an innovative game designed first and foremost for gamers who enjoy strategic tower defense experiences, with educational value as a natural byproduct. While our primary audience is gamers seeking challenging and engaging gameplay, educators, teachers, and students interested in quantum computing concepts represent a valuable secondary audience.

Our approach prioritizes fun gameplay (60%) over educational content (40%), ensuring that The Qubit succeeds as an entertaining game first, with learning as a beneficial side effect rather than the primary focus. Taking inspiration from games like Elden Ring that respect players' intelligence and learning capacity, The Qubit assumes players can think critically and learn through experience. We're creating a game that's both fun and challenging, never talking down to players or oversimplifying for the sake of accessibility. Instead, the game allows players to experiment, fail, and develop mastery through iterative gameplay.

Players protect a "Quantum Core" in this game from waves of enemies that stand in for decoherence and other quantum dangers. The ideas of quantum computing are explicitly reflected in the gameplay mechanics:

- On a grid, players arrange several qubit kinds ($|0\rangle$ and $|1\rangle$), each of which has a unique function.
- Red $|1\rangle$ qubits attack enemies, whereas blue $|0\rangle$ qubits generate resources.
- Quantum gates manipulate qubit states by acting as momentary area effects.
- Qubits are connected by the entanglement mechanism to improve strategic impact.
- Real-world problems with quantum computing are represented by specialised adversaries, such as "observer enemies" that push qubits out of superposition.

These mechanics will be progressively developed throughout the project to ensure both strategic depth and quantum computing accuracy. The game's difficulty curve will be carefully balanced to provide an appropriate challenge that encourages learning through trial and error rather than explicit tutorials. Through these engaging mechanics, players naturally absorb quantum computing fundamentals while enjoying compelling strategic gameplay.

Project Motivation

With big businesses like Google, IBM, and Microsoft heavily investing in research and development, quantum computing is poised to transform computing technology (Tran, 2025). Nonetheless, there is a substantial knowledge deficit in the field:

- **Learning barrier:** The concepts of quantum physics are paradoxical and typically taught using sophisticated mathematics, which makes them difficult to understand.
- **Growing demand for quantum literacy:** Professionals who comprehend these ideas are becoming more and more necessary as quantum technologies develop, yet conventional educational materials are not enough.
- **Need for intuitive learning methods:** Abstract topics are better understood through interactive, hands-on learning methods than through passive theoretical study.
- **Gamification benefits:** According to research, gamification enhances learning results in STEM subjects, especially for intricate subjects that gain from interaction and visualisation (Playfoot, 2016).

The Qubit directly addresses these challenges by providing:

- An interactive platform where users experiment with quantum principles firsthand
- Visual representations that make abstract concepts concrete
- Strategic gameplay that reinforces understanding through application
- A low-pressure environment to fail, learn, and internalize counterintuitive concepts

User Story, Context, and Beneficiaries

Quantum computing represents a paradigm shift in computational power, but remains inaccessible to many due to its complex nature. Traditional learning resources often rely on advanced mathematics and theoretical physics, limiting who can engage with this emerging field. Our game creates a bridge to quantum understanding for various groups:

- **Gamers:** Our primary audience who will enjoy a strategically deep tower defense game that introduces quantum concepts through engaging gameplay rather than explicit education
- **Students:** Gain hands-on understanding of quantum computing concepts without requiring an advanced mathematics background
- **Educators:** Access an interactive teaching tool to visualize and demonstrate quantum principles
- **Enthusiasts:** Explore an accessible introduction to quantum computing concepts in an engaging format
- **Researchers:** Find inspiration for presenting complex concepts in more intuitive ways

By creating a fun, visually engaging experience, The Qubit opens doors to quantum computing for diverse audiences who might otherwise be intimidated by the field's complexity.

Team description

Our team structure leverages each member's unique strengths:

- **Hermanto**: Lead developer responsible for core game systems due to extensive Unity experience
- **HaoYu Zhang**: Focused on UI development, including the main menu system
- **Tiantian Wang**: Supporting development and document management, coordinates team meetings
- **Ze Kai Chong**: Unity developer and on-campus liaison for updating the team on lecture information
- **Mohan Dash Tanu**: Quantum computing subject matter expert, ensuring accurate implementation of quantum concepts
- **Jiali Gu**: Graphic designer responsible for visual assets and game aesthetics

Document Organisation

Project documentation is centrally stored in our GitHub repository within the 00_Documentation folder, organized into four main subdirectories:

- **DesignDocs**: Contains key design-related files, including the initial development checklist and core design document
- **GeneralDocs**: Houses major project reports such as pitch presentations, business plans, and milestone reports
- **MeetingMinutes**: Structured by week, with templated documentation for both client and internal meetings
- **Timesheets**: Individual teammate contributions are tracked weekly

This organization ensures all documents remain consistent, accessible to each team member, and properly version-controlled.

Communication & Project Management

- **Primary platform**: Discord with specialized channels for resources, bug reports, task assignments, code reviews, and meeting planning
- **Progress tracking**: Bi-weekly progress reports coinciding with supervisor meetings (Wednesday) and internal team meetings (Friday)

- **Risk management:** Cross-training ensures multiple team members can address each system component if issues arise
- **Documentation:** Centralized in GitHub repository with consistent structure, formatting, and naming conventions

Implementation Strategy

Throughout all milestones, we'll maintain our core 60/40 balance, favoring gameplay over educational content. Our approach follows these principles:

1. **Vertical Slices:** Each development phase will produce a playable version of the game with increasing complexity
2. **Test-Driven Development:** Regular playtesting will guide our design decisions
3. **Iterative Implementation:** Systems will be built, tested, and refined throughout the development cycle
4. **Core First:** We'll ensure the tower defense fundamentals are solid before layering quantum mechanics
5. **Player Experience Priority:** All educational elements must enhance rather than detract from gameplay

This approach ensures we create a game that's both entertaining and educational, with clear development milestones to track progress and manage scope throughout the project.

References

Playfoot, J., 2016. Exploring the role of gamification within STEM teaching as a mechanism to promote student engagement, develop skills and ultimately improve learning outcomes for all types of students. In *EDULEARN16 Proceedings* (pp. 2140-2147). IATED.

Tran, B. (2025) Big Tech's Quantum Computing Investments: Google, IBM, and Microsoft by the numbers, PatentPC. Available at: <https://patentpc.com/blog/big-techs-quantum-computing-investments-google-ibm-and-microsoft-by-the-numbers> (Accessed: 02 April 2025).

MCI project First Milestone Plan

Team: KM5A

Project Title: Quantum Computing Game Based

Milestone 1	Activities	Projected Outputs
Core Tower Defense Mechanics	Implement grid system with visual feedback	Define projected outputs from your work plan
	Create Basic Qubit Placement System with Both Types ($ 0\rangle$ and $ 1\rangle$)	Working placement system allowing players to select and place different qubit types ($ 0\rangle$ and $ 1\rangle$) with distinct visual representations and behaviours
	Implement enemy path system and basic enemy AI	Enemies that follow paths and attack qubits/core with appropriate behaviours
	Develop wave management system with progressive difficulty	Working wave system that spawns different enemy types with increasing difficulty
	Build resource generation and management system	Resource UI and economy system that allows for placing and upgrading qubits
	Implement win/lose conditions and game restart functionality	Game loop that properly detects win/loss and allows for restart
	Create basic UI for game status, qubit selection, and resources	Complete UI system that communicates all necessary game information

MCI project Second Milestone Plan

Team: KM5A

Project Title: Quantum Computing Game Based

Milestone 2	Activities	Projected Outputs
Quantum Mechanics Layer	Implement quantum gates as special abilities/consumables (X Gate, H Gate)	• Functioning gate mechanics that can be applied to qubits during gameplay • Visual effects showing gate application and state changes • UI elements for gate selection and feedback on effects
	Create superposition mechanic allowing qubits to exist in multiple states	• Qubits that can operate in superposition states with distinct visual indicators • Probabilistic effects system reflecting quantum measurement principles • Documentation explaining the superposition implementation and its relation to quantum concepts
	Develop entanglement system linking qubits for enhanced effects	• Visual representation of entanglement connections between qubits • Functional entanglement mechanics affecting linked qubit behaviors • Strategic gameplay options enabled by entanglement properties
	Design and implement quantum-specific enemy types	• Enemy behavior patterns that require strategic use of quantum mechanics to counter • Observer Enemies that collapse superposition states • Thermal Noise Enemies that introduce randomness to quantum systems
	Create quantum-themed visual effects to represent quantum states and interactions	• Distinctive visual language for different quantum states ($0\rangle$, $1\rangle$, superposition) • Animation system for state transitions and measurements • Particle effects for quantum interactions and gate applications